Gravity as the Large-Scale Emergence of Dual Attractive Forces: 3D Magnetism and Thermal Gradients

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Abstract

This experimental physics paper refutes General Relativity (GR) and Thermodynamics by empirical tests that contradicted its predictions. We demonstrate that GR's foundational dependency, Maxwell's Equations, fails in three-dimensional systems where magnetic repulsion is geometrically impossible and we provide empirical evidence that heat is a vectorial attractive force capable of generating negentropy.

Our experiments reveal that "Gravity" is the macroscopic manifestation of two distinct attractive forces: (1) 3D Magnetic Alignment, which governs solid bodies and precludes stable repulsion, and (2) Thermal Attraction, demonstrated by the "antigravitational" flow of fluids toward heat sources. We present a new cosmological framework where elliptical orbits are resolved not as a product of curved spacetime, but as the dynamic equilibrium between magnetic alignment (structural lock) and thermal attraction (orbital pull).

1. Introduction: The Hierarchical Structure of Modern Physics:

Modern physics is hierarchical, with each theory built upon the framework beneath it:

Maxwell (1865) → Special Relativity (1905) → General Relativity (1915).

Einstein acknowledged that Special Relativity owes its origins to Maxwell's equations. Consequently, if Maxwell's framework fails empirically in three-dimensional systems, the justification for relativity evaporates. This work demonstrates that failure.

2. Maxwell's Framework and Einstein's Dependency: Maxwell's Equations treat magnetic attraction and repulsion as perfectly symmetric:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

The curl operator predicts a 1:1 ratio between attraction and repulsion in free systems. Einstein abandoned absolute time and space (SR) specifically to preserve this framework when faced with contradictions in Newtonian mechanics. If Maxwell fails, the hierarchical tower of SR and GR collapses.

3. Experiment 1: Rectangular Magnets and the 7:1 Asymmetry: Testing two N52 neodymium magnets in 64 configurations revealed a fundamental asymmetry:

Outcome	Maxwell Prediction	Observed
Direct Attraction	32 (50%)	16 (25%)
Rotational Attraction	0 (0%)	40 (62.5%)
Sustained Repulsion	32 (50%)	8 (12.5%)

Ratio: Attraction : Repulsion = 7 : 1.

Repulsion exists only as a transient corrective mechanism (torque: $\tau = m \times B$) inducing rotation toward attraction.

4. Experiment 2: Spherical Magnets: In spherical systems with full 3D rotational freedom, stable repulsion is geometrically impossible (Video S1). Spheres oscillate until they snap into attraction. Cosmic bodies, being spherical and frictionless, must resolve every magnetic interaction into attraction or orbital equilibrium.

5. The Dimensional Distinction:

Quantum Mechanics (2D): Operates in planar/constrained regimes (orbitals, detector planes) where full 3D torque cannot operate, allowing Maxwell's symmetry to hold.

General Relativity (3D): Attempts to apply the broken Maxwellian symmetry to volumetric 3D reality requires the "fictional" geometrization of spacetime to compensate.

6. Thermal Attraction and the Post-Einsteinian Reality

6.1 Lipid Demonstration experiment: using an inclined plane and a heat source beneath it (Video S2) show matter (butter) climbing upward against gravity toward the heat source. This confirms that an active thermal gradient creates a directional attractive force.

6.1.1 The "Iceberg" Experiment: Refutation of the Second Law of Thermodynamics

To rigorously test the interaction between thermal gradients and gravity, we designed the "Iceberg Experiment" (Video S3). A stainless-steel pan containing water was inclined and placed in a freezer, fixing a solid ice block at the lowest gravitational potential. Upon removal, a ping-pong ball was used to mechanically verify the negative gradient (downward slope) before and after the experiment to rule out deformation or levelling errors.

The pan was then placed on a stove with a high-intensity flame directed at the highest point of the incline. The experiment ran for over 18 minutes (The original video has been sped up by 5.0x). Observations:

1. **Gravitational Baseline:** Initial meltwater obeyed standard mechanics, flowing downward.
2. **Thermal Reversal (Negentropy):** As the thermal gradient intensified, the fluid flow reversed. Water migrated *upward* against the gravitational vector, effectively "climbing" the slope to reach the heat source.

3. **Thermodynamic Popper's Falsification:** The system maintained a stable dual-state: violent boiling occurred at the peak (heat source) while the ice block remained solid at the base, separated by only 15 centimeters and always showing continuity between the frozen and boiling water.
4. **Marangoni effect cannot explain this upward migration:** According to surface tension gradients, fluids flow from regions of low surface tension (high temperature) to regions of high surface tension (low temperature). In our setup, the Marangoni vector is directed downward, acting in conjunction with gravity to push the fluid away from the heat source. The fact that the fluid migrates upward confirms the existence of a third, dominant vector—**Thermal Attraction**—capable of overcoming both gravitational potential and the thermocapillary Marangoni flow.

Conclusion: this behavior violates the Second Law of Thermodynamics. Rather than dissipating into uniform thermal equilibrium (entropy increase), the thermal gradient actively organized matter against gravity. The heat source functioned as an attractive ordering force, creating a low-entropy flow that overpowered gravitational potential.

6.2: Leidenfrost Orbits and the Elliptical Resolution

As water droplets reached the peak thermal gradient, they did not evaporate instantly but entered stable orbital patterns (Leidenfrost effect). This behavior models the macroscopic mechanics of cosmic systems.

We propose that **cosmological orbits are elliptical** because they are the product of two competing attractive forces, not a single gravitational curve:

1. **3D Magnetic Alignment:** This force seeks rigid structural locking and alignment (as seen in the spherical magnet experiments).
2. **Thermal Attraction:** This force pulls the orbiting body toward the star (the heat source).

An elliptical orbit is the "breathing" pattern of a celestial body oscillating between the magnetic impulse to align (torque) and the thermal impulse to merge (attraction). This eliminates the need for "initial velocity" assumptions or "Dark Matter" halos to explain galactic rotation curves.

7. Conclusion: Hierarchical Collapse

The logic is irrefutable, and the evidence is empirical:

1. **Maxwell fails in 3D**, making spherical repulsion impossible and mandating universal aggregation.
2. **The Second Law of Thermodynamics is refuted**, as heat acts as a negentropic, attractive force that can order matter against gravity.
3. **General Relativity is obsolete**, as it fails to account for the thermal vector in gravity and relies on a broken magnetic symmetry.

The universe requires no deep imaginative "spacetime curvature", "dark matter", or "dark energy". It requires only 3D magnetic alignment, thermal attraction, and rotational freedom.

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